

Client REQUEST-body wire-shape audit — clients/ota-manager

Revision: 1 Last modified: 2026-07-10T00:00:00Z

Scope

Follow-up to the prior response-shape reconciliation pass. This pass audits every REQUEST-body interface (*Request / *Payload / create-update-recall input shapes) declared in clients/ota-manager/src/lib/api-client.ts against the real Go request struct the corresponding server/internal/api/*.go handler decodes, and fixes every caller (adapter-shim hooks, dialogs/forms) that builds a request body from a drifted interface.

Method per §11.4.6 (no-guessing) / §11.4.108 (runtime-signature) / §11.4.115 (RED-baseline-on-broken-artifact): every claim below cites the exact server file:line for the real Go struct. Nothing is asserted without a citation.

Per-interface drift table

Client interface	Verdict	Drift	Server citation
LoginRequest	FIXED	Client sent {email, password}; server requires {username, password}. Every login would 400.	server/internal/wire.go:19-22 (LoginRequest), handlers_auth.go (handleLogin)
DeviceRegistrationRequest	FIXED (unused)	Client shape {device_id, board, firmware_version, hardware_revision, serial_number} matched none of the real fields. Discovered	server/internal/wire.go:41-49 (DeviceRegistrationRequest)

Client interface	Verdict	Drift	Server citation
CreateReleaseRequest	FIXED	unused — use- devices.ts's useRegisterDevice already used the correct DeviceRegistration type, so this had zero runtime impact; fixed for wire-shape correctness of the exported type. Client sent {project_id, version, file_url, file_hash, changelog, target_board, firmware_version}; server requires {artifact_id, version, os, target_model, notes?, min_current_version?}. Real required artifact_id/os/ target_model were entirely absent. Client sent {deployment_id, groups, rollout_percentage?, staged}; server requires {phases: RolloutPhaseSpec[]} only (deployment id is a URL param). Real required phases was entirely absent — every create-rollout submission would 400 (len(req.Phases) == 0).	server/internal wire.go:137-144 (ReleaseCreate)
CreateRolloutRequest	FIXED	Client sent {reason, force}; server requires	server/internal handlers_rollou (RolloutCreate), (handleCreateRo
RecallRequest		Client sent {reason, force}; server requires	server/internal handlers_recall

Client interface	Verdict	Drift	Server citation
	FIXED (RED → GREEN proof below)	{to_release_id, reason?}. force does not exist server-side; to_release_id (mandatory) was entirely absent — the recall endpoint rejected EVERY request this client sent. Client sent {release_id, group_ids: string[], strategy, rollout_percentage?, staged}; server	(RecallRequest), (handleRecall)
CreateDeploymentRequest	FIXED	requires {release_id, strategy, group?} — a single optional group name, never an array, and no rollout_percentage/staged field on this request at all. Client sent {name, description, device_ids: string[], labels: Record<string, string>; server requires {name, description?} only. device_ids/labels do	server/internal wire.go:167-171 (DeploymentCrea
CreateGroupRequest	FIXED	not exist on group creation — Go's JSON unmarshal silently ignored them (silent-data-loss bug, not a 400: the operator believed a new group started with the given members/labels; the server always created an empty group).	server/internal handlers_group. (GroupCreate)

Client interface	Verdict	Drift	Server citation
UpdateGroupRequest	FIXED	Client carried an extra labels? field; groups have no labels concept anywhere on the server. Silently ignored (same silent-drop class as above).	server/internal_handlers_group. (GroupUpdate)
AddGroupMembersRequest	VERIFIED CORRECT	{device_ids: string[]} — exact match, no drift.	server/internal_handlers_group. (MemberAdd)
ArtifactUploadMetadata	VERIFIED CORRECT	Field-for-field match (sha256, signature, version, os, target_model, file_hash?, file_size?, metadata_hash?, metadata_size?, payload_offset?, payload_size?), no drift.	server/internal_wire.go:107-119 (ArtifactUpload)
DeltaRegisterRequest	VERIFIED CORRECT	{base_artifact_id, target_artifact_id, sha256?, size?, storage_ref?} — exact match, no drift.	server/internal_handlers_delta. (DeltaRegister)
DeltaFindParams	VERIFIED CORRECT (out of scope)	Query-string params (?base=&target=), not a JSON body — outside this audit's strict scope, but verified correct against c.Query("base")/c.Query("target") while auditing its sibling.	server/internal_handlers_delta. (handleFindDelta)
RefreshRequest (inline, no named client interface)	VERIFIED CORRECT	use-auth.ts's useRefresh already sends {refresh_token} inline, matching the	server/internal_wire.go:25-27 (RefreshRequest)

Client interface	Verdict	Drift	Server citation
		server exactly. No interface to fix.	

Discovered but explicitly NOT fixed (out of strict wire-shape scope)

- **CreateProjectRequest/UpdateProjectRequest** (server/internal/api/wire.go:248-257) have no corresponding client interface in api-client.ts at all, and no page/hook in clients/ota-manager currently builds a project create/update request — project management is unwired in this SPA. Adding brand-new, never-exercised interfaces was judged out of scope for a drift-reconciliation pass (would be fabricating untested surface, not fixing existing drift).
- **CreateDeploymentRequest.strategy value domain:**
handleCreateDeployment (server/internal/api/handlers_deployment.go:39-42) accepts ONLY the literal "all-targets" for this MVP. The useCreateDeployment.ts dialog's strategy picker (rolling/blue-green/canary/full) maps to rolling/blue_green/canary — all of which are still rejected server-side with a 400. This is a **business-rule/value-domain** mismatch (the Go field is untyped string), not a JSON structural drift, so the wire *shape* is now correct but the deployment-creation flow will still 400 until the strategy picker is redesigned for the MVP's real capability. Documented in-code (useCreateDeployment.ts).
- **CreateRolloutRequest phase-plan validation:** the rollout-engine brick validates phase plans as strictly-increasing percentages ending at 100 (handlers_rollout.go:80-81 comment). useEvaluateRollout.ts's single-percentage panel now builds a technically-correct {phases: [...]} wire shape, but a submission below 100% may still be rejected as an invalid phase plan — a real phase-plan builder UI is a separate, out-of-scope work item. Documented in-code.

Component/hook changes required by the corrected interfaces

File	Change
src/hooks/use-auth.ts	useLogin now maps its UI-facing LoginCredentials.email onto the wire LoginRequest.username field

File	Change
	at the request boundary (login form UX unchanged).
src/hooks/ useCreateDeployment.ts	Drops fabricated group_ids/ rollout_percentage/staged; builds {release_id, strategy, group?}.
src/hooks/useCreateRelease.ts	Drops fabricated project_id/ file_url/file_hash/changelog/ target_board/firmware_version; builds {artifact_id, version, os, target_model, notes?, min_current_version?}.
src/hooks/useEvaluateRollout.ts	Drops fabricated deployment_id/ groups/rollout_percentage/ staged; builds {phases: [RolloutPhaseSpec]} from the panel's single percentage value.
src/hooks/useRecall.ts	RecallInput gains a required toReleaseId field (was force?: boolean, dropped); builds {to_release_id, reason?}.
src/hooks/useCreateGroup.ts	Drops fabricated device_ids: []/ labels: {}; builds {name, description?}.
src/features/deployments/ deployment-detail-page.tsx	RecallSection gains a new required “Target release ID” input (recallToReleaseId state, threaded via new props) — previously there was no way for the UI to supply the server's mandatory to_release_id; the Recall button is now gated on this field being non-empty (previously gated only on the reason field).
src/lib/api-client.ts	8 request-body interfaces fixed (LoginRequest, DeviceRegistrationRequest, CreateReleaseRequest, CreateRolloutRequest, RecallRequest,

File	Change
src/__tests__/wire-shape-request-body.test.ts	CreateDeploymentRequest, CreateGroupRequest, UpdateGroupRequest); 3 verified correct with citation comments added (AddGroupMembersRequest, ArtifactUploadMetadata, DeltaRegisterRequest); 1 verified-correct-but-out-of-scope noted (DeltaFindParams). New compile-time-only guard test (tsc --noEmit oracle), mirroring the existing wire-shape-pagination.test.ts/wire-shape-nonlist.test.ts pattern — asserts every corrected interface against a real-server-shaped fixture and pins every enumerated OLD fabricated shape as a permanent @ts-expect-error.

No response interfaces were touched (ReleaseResponse, Deployment, DeploymentStatus, RolloutState, RolloutDecision, RollbackView, Group, DeploymentResponse, etc. — untouched, per strict scope).
DeploymentResponse (a fabricated response shape: {id, group_ids, rollout_percentage, staged, created_by}) is confirmed unused by any live caller (grep: only declared + re-exported) — left as-is, out of scope.

§11.4.115 RED → GREEN proof (RecallRequest)

RED (drift reintroduced, real Go struct unchanged)

RecallRequest in src/lib/api-client.ts was temporarily reverted to the OLD fabricated shape while src/hooks/useRecall.ts kept the corrected {to_release_id, reason} body construction:

```
// TEMPORARILY REINTRODUCED for RED capture:
export interface RecallRequest {
  reason: string;
```

```
    force: boolean;
}
```

Command: `npx tsc --noEmit`

Real captured output:

```
src/hooks/useRecall.ts(26,9): error TS2322: Type 'string |
undefined' is not assignable to type 'string'.
    Type 'undefined' is not assignable to type 'string'.
```

This proves the fix is load-bearing: the corrected `useRecall.ts` body (`to_release_id`, `reason`) is structurally incompatible with the old drifted interface — exactly the defect class this audit closes (the real server would have rejected every recall the OLD client shape sent, per `handleRecall`'s `req.ToReleaseID == ""` check, `server/internal/api/handlers_recall.go:68-72`).

GREEN (fix restored)

`RecallRequest` restored to:

```
export interface RecallRequest {
  to_release_id: string;
  reason?: string;
}
```

Command: `npx tsc --noEmit`

Real captured output: **(empty — exit 0)**

Final verification (post-fix, full suite)

```
$ npx tsc --noEmit
(no output — exit 0)
```

```
$ npx vitest run
RUN v4.1.9 /home/milos/Factory/projects/tools_and_research/
helix_ota/clients/ota-manager
```

```
Test Files  14 passed (14)
Tests       52 passed (52)
Start at    04:36:47
```


Duration 9.23s (transform 273ms, setup 417ms, import 1.31s, tests 943ms, environment 5.38s)

Files changed

- clients/ota-manager/src/lib/api-client.ts
- clients/ota-manager/src/hooks/use-auth.ts
- clients/ota-manager/src/hooks/useCreateDeployment.ts
- clients/ota-manager/src/hooks/useCreateRelease.ts
- clients/ota-manager/src/hooks/useEvaluateRollout.ts
- clients/ota-manager/src/hooks/useRecall.ts
- clients/ota-manager/src/hooks/useCreateGroup.ts
- clients/ota-manager/src/features/deployments/deployment-detail-page.tsx
- clients/ota-manager/src/__tests__/wire-shape-request-body.test.ts (new)